MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548046

(COUNTER, RANGE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range counter used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specifications for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548046	Range Counter
F10548009	PRF Counter

PACKAGING DATA SHEET

10548046	Packaging of Range Counter

(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548046 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the card assembly.

3.5 Performance. - For the requirements of 3.5.4 through 3.5.7, logic input and output levels shall be as follows:

Logic 0: 0.0 ±0.5 volt direct current (Vdc) level.
Logic 1: +3.5 ±1.0 Vdc level.

3.5.1 Input power. - The input power applied between terminals P1-30, P1-31 and P1-32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ±0.3 Vdc, at 200 milliamperes (mA), regulated; with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Clock signals. - With a 7.5 ±0.5 megahertz (MHz) square wave or sine wave applied to P1-43 there shall be a 7.5 MHz clock signal at P1-45 and P1-47. The input signal at P1-43 shall have an amplitude of 5.0 ±0.2 volts into a 50-ohm load. The clock signal at P1-47 (7.5 MHz Cp) shall have an amplitude of 3.4 ±1.0 Vdc and a period of 133 ±10 nanoseconds (ns). The delayed clock signal at P1-45 (7.5 MHz Cpd) shall have an amplitude of 3.5 ±1.0 Vdc, a period of 133 ±10 ns, and be delayed 20 ±10 ns from the signal at P1-47.

3.5.3 Input signals. - The input signals to the range counter shall be the signals $\overline{\text{PRF}}$ RESET, $\overline{\text{PRF}}$, T1, $\overline{\text{T1}}$, $\overline{\text{T2}}$, T3, and STROBE as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 V to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±1.0 V and return to +3.5 V.

3.5.4 MIN RANGE INHIBIT output. - With an input logic 0 level applied to P1-1 there shall be a logic 1 level present at P1-14. With an input logic 1 level applied to P1-1 the MIN RANGE INHIBIT output signal as shown in Figure 1 shall be present at P1-14.

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3.5.5 E/L COURSE output. - With a logic 1 applied to P1-4 and the input logic levels applied sequentially as specified in Table I, the E/L course output signals at P1-51 shall be shown in Figure 1 with positive pulse durations as specified in Table I. With a logic "0" applied to P1-4, the E/L course output signal at P1-51 shall be a logic "0".

TABLE I. - OUTPUT SIGNALS

	INPUT TERMINALS									OUTPUT SIGNAL DURATIONS	
STEP	12	56	21	33	35	38	40	48	55	51 TERMINALS	49
1	1	0	1	0	0	0	0	0	1	0.28 ±0.05 μs	2.65 ±0.15 μs
2	0	0	1	0	0	0	0	0	1	0.41 ±0.05 μs	2.80 ±0.20 μs
3	1	1	1	0	0	0	0	0	1	0.53 ±0.08 μs	3.05 ±0.25 μs
4	1	0	0	0	0	0	0	0	1	0.82 ±0.10 μs	3.40 ±0.20 μs
5	1	0	1	1	0	0	0	0	1	1.30 ±0.15 μs	3.80 ±0.20 μs
6	1	0	1	0	1	0	0	0	1	2.40 ±0.20 μs	4.85 ±0.35 μs
7	1	0	1	0	0	1	0	0	1	4.60 ±0.60 μs	7.05 ±0.35 μs
8	1	0	1	0	0	0	1	0	1	8.58 ±0.58 μs	11.6 ±0.8 μs
9	1	0	1	0	0	0	0	1	1	17.0 ± 1.0 μs	19.6 ±0.8 μs

3.5.6 PRE RANGE TRIG output. - With the input logic levels applied sequentially as specified in Table I, the PRE RANGE TRIG output signal at P1-49 shall be as shown in Figure 1 with pulse duration as specified in Table I. There shall be a delay of 1.7 ±0.1 microsecond (μs) between the leading edge of the PRE RANGE TRIG output signal at P1-49 and the leading edge of the E/L COURSE output signal at P1-51 for all steps of Table I.

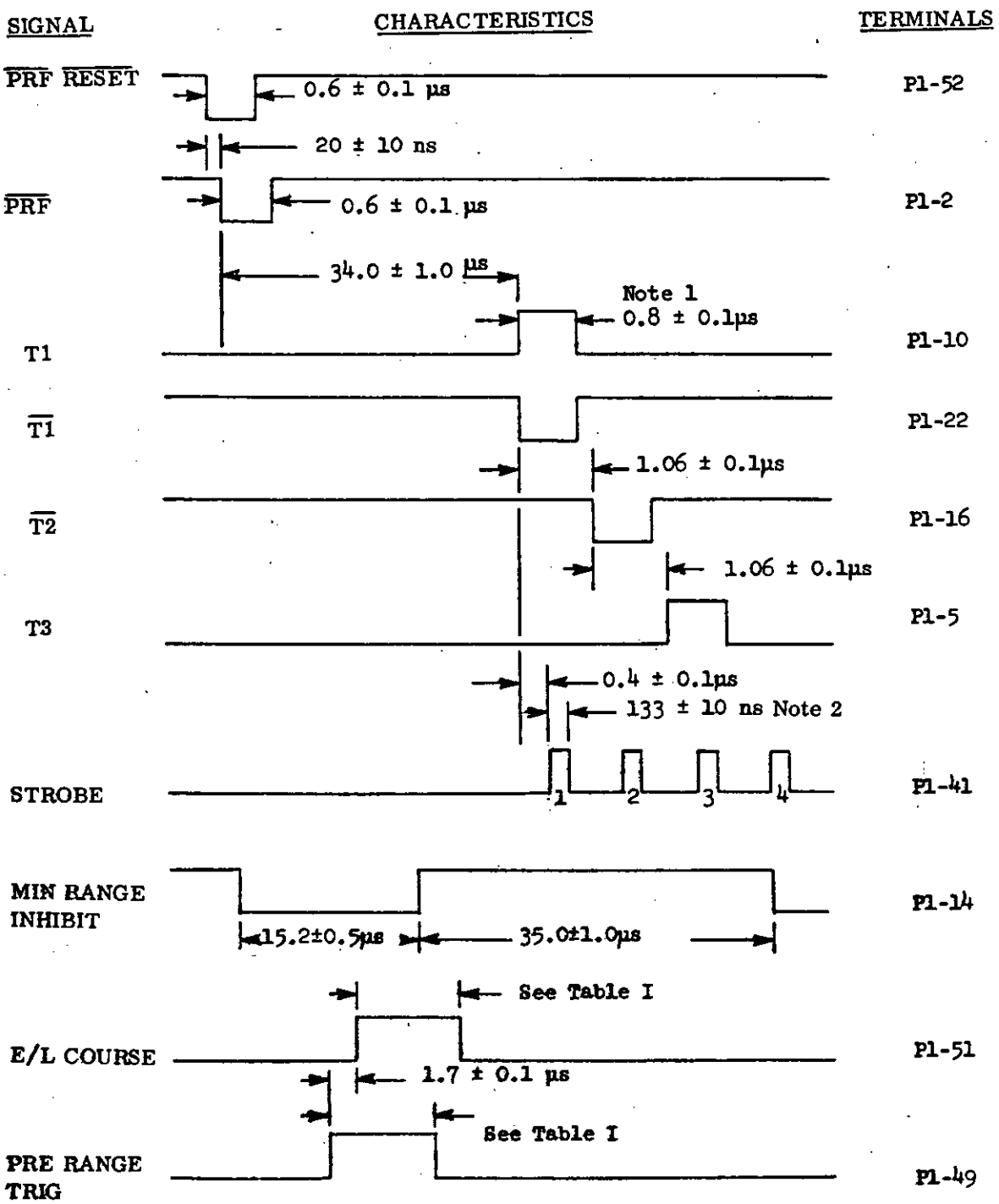
3.5.7 T2 output. - At P1-46 the T2 output signal shall be present. The T2 signal shall be complementary to and have the same characteristics as $\overline{T2}$ shown in Figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

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NOTES: 1. All T pulses 0.8 ±0.1 µs wide.
2. STROBE signal is a train of 15 pulses each 133 ±10 ns wide.
3. All signals at a 20-kHz PRF.

FIGURE 1. INPUT AND OUTPUT SIGNALS

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4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test Equipment. - In conjunction with 4.5 the following test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.3.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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TABLE II. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak. Capacity: 400 mA.
2. Signal generator	7.5 ±0.5 MHz square wave or sine wave. Amplitude: 5.0 ±0.2 V peak-to-peak into 50 ohms.
3. Oscilloscope	Dual channel: Rise time of scope plus measuring probe; 10 ns or less. Sweep period: 50 ns per cm or greater. Input resistance: 8 megohms or greater.
4. Test fixture	As specified in Figure 2. See Table III for test fixture parts list.

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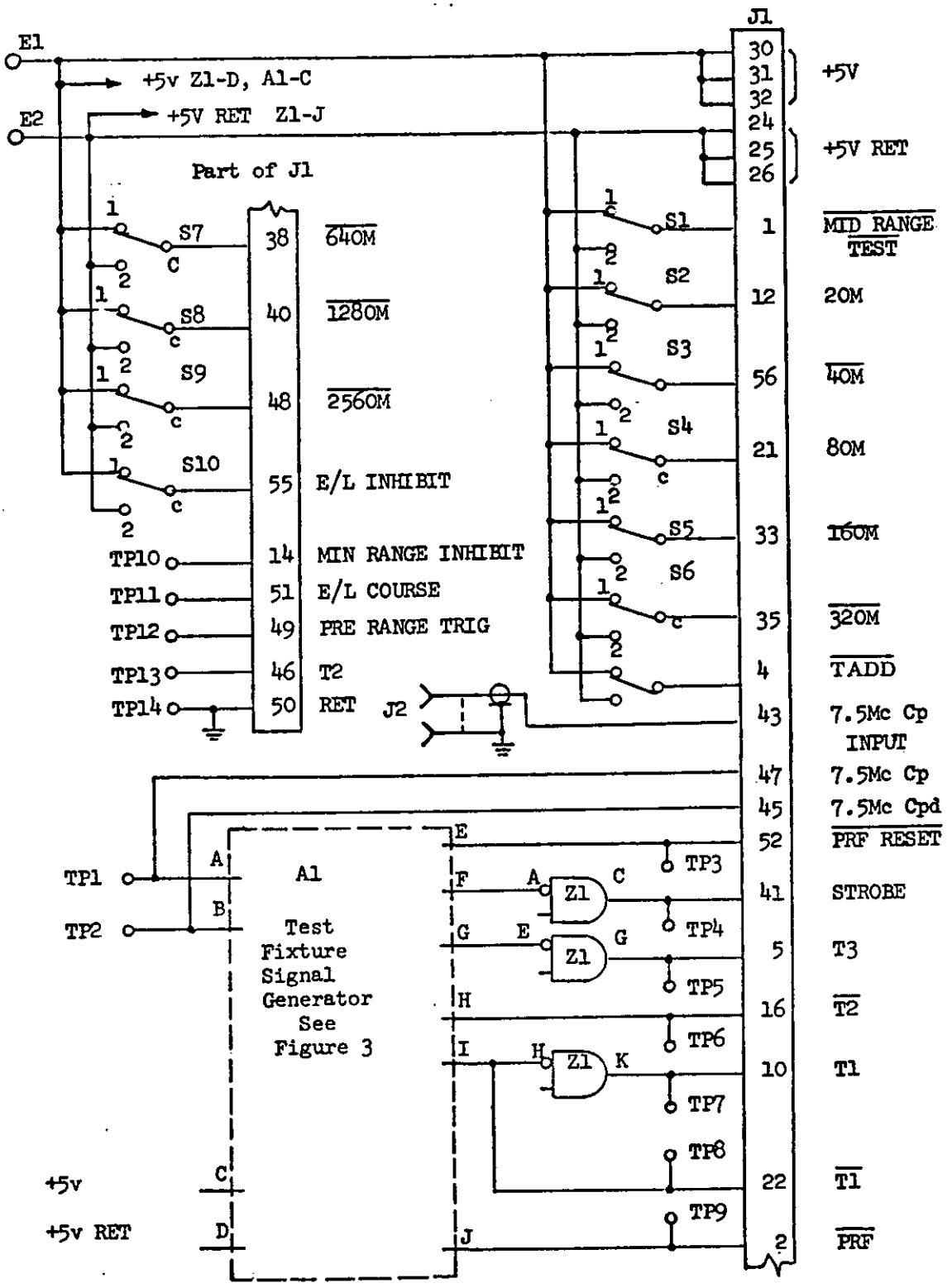


FIGURE 2. TEST FIXTURE SETUP

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TABLE III. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Test fixture signal generator in accordance with Figure 3. See Table IV for parts list, or applicable circuits on a PRF counter assembly, F10548009.
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
S1 thru S11	Switch SPDT.
TP1 thru TP14	Tip jack, H. H. Smith 1501 or equivalent.
Z1	Integrated circuit, SG-140-03 Sylvania or equivalent.

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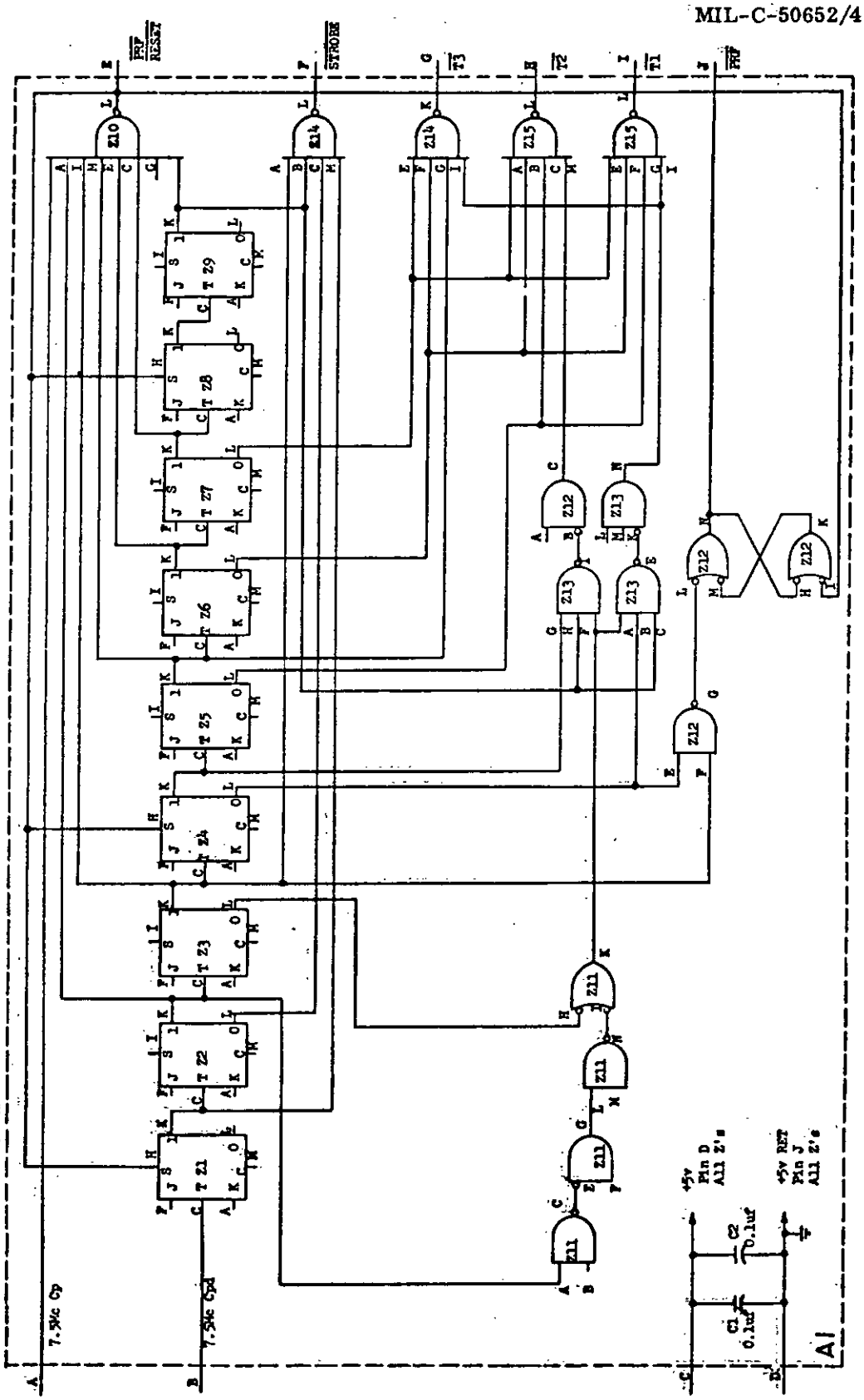


FIGURE 3. TEST FIXTURE SIGNAL GENERATOR

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TABLE IV. - TEST FIXTURE SIGNAL GENERATOR PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
C1	Capacitor 0.1 μF at 50 Vdc, SSM-1-189 Gulton Industries, Inc. or equivalent.
C2	Same as C1
Z1 thru Z9	Integrated circuit, SF-50-28 Sylvania or equivalent.
Z10	Integrated circuit, SG-160-21 Sylvania or equivalent.
Z11	Integrated circuit, SG-140-37 Sylvania or equivalent.
Z12	Same as Z11.
Z13	Integrated circuit, SG-190-21 Sylvania or equivalent.
Z14	Integrated circuit, SG-40-28 Sylvania or Equivalent.
Z15	Same as Z14.

4.6.3 Performance. - The selected range counter circuit card assemblies shall be tested as follows: Set up equipment as shown in Figure 4. Use oscilloscope, item 3, for observation of input and output signals specified in 4.6.3.2 through 4.6.3.7. When specified, a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a +3.5 ±1.0 Vdc level. Switches S1 through S10 on test fixture, item 4, are used to supply the logic level input signals specified in 4.6.3.4 through 4.6.3.7. Switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input power. - Using item 1, apply input power as specified in 3.5.1.

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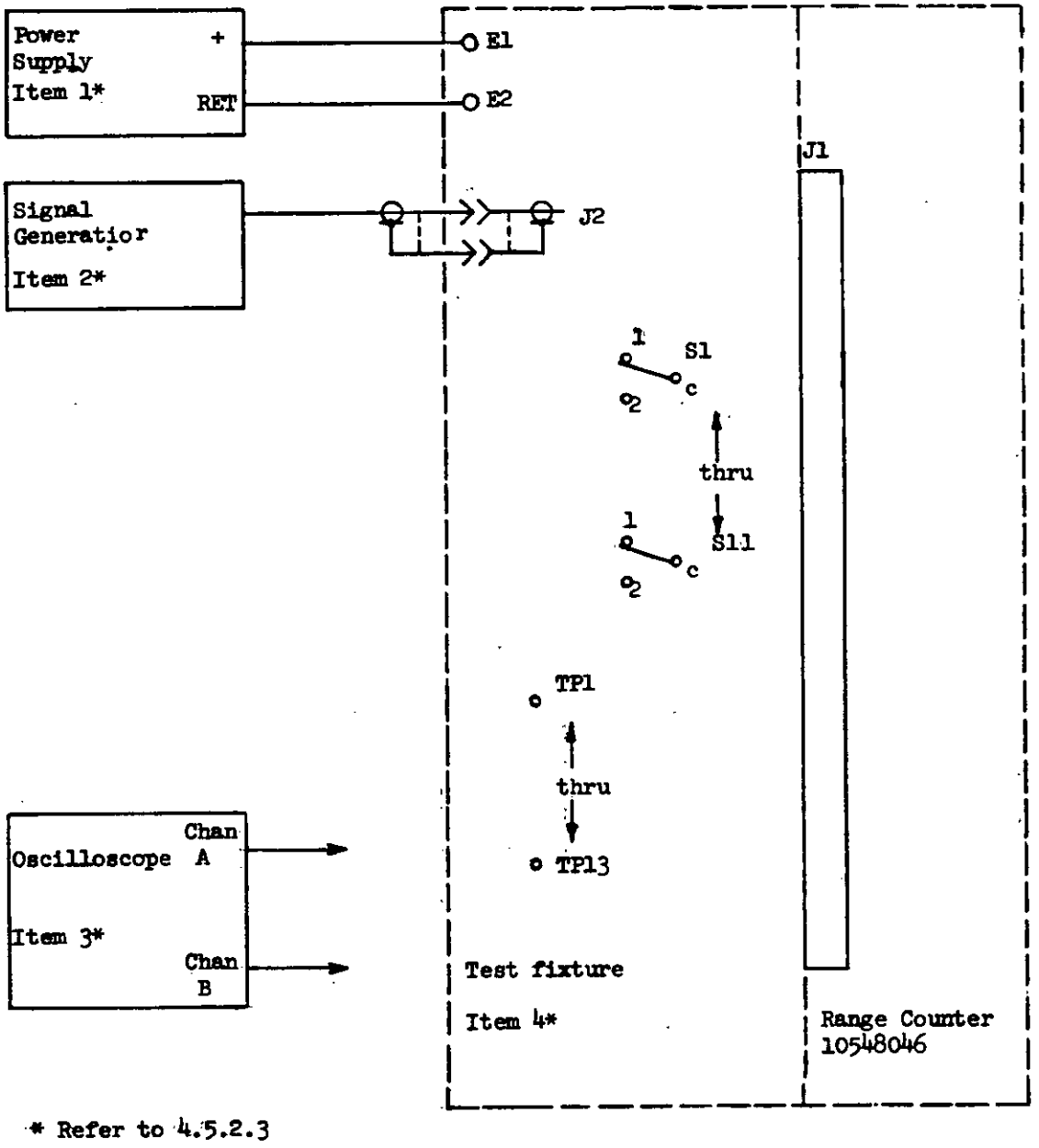


FIGURE 4. RANGE COUNTER TEST SETUP.

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4.6.3.2 Clock signals. - Using item 2, apply a 7.5 ±0.5 megahertz (MHz) signal to J2 and with oscilloscope, item 3, observe clock output signals Cp and Cpd at TP1 and TP2 respectively to determine compliance with 3.5.2.

4.6.3.3 Input signals. - Using item 3, verify input signals $\overline{\text{PRF RESET}}$, STROBE, T3, $\overline{\text{T2}}$, T1, $\overline{\text{T1}}$, and $\overline{\text{PRF}}$ at TP3 through TP9 respectively for compliance with Figure 1. The input signals are generated on test fixture subassembly A1, as shown in Figure 3.

4.6.3.4 MIN RANGE INHIBIT output. - Use switch S1 to apply logic levels as specified in 3.5.4. At TP10, observe the MIN RANGE INHIBIT output signal as specified in 3.5.4.

4.6.3.5 E/L COURSE output. - Use switches S2 through S10 to apply logic levels, as specified in Table 1, to terminals P1-12, P1-56, P1-21, P1-33, P1-35, P1-38, P1-40, P1-48, and P1-55 respectively. Use switch S11 to apply logic levels as specified in 3.5.5 to P1-4. At TP11, observe the E/L COURSE output signal as specified in 3.5.5.

4.6.3.6 PRE RANGE TRIG output. - Use switches S2 through S10 as specified in 4.6.3.5 to apply logic levels as specified in 3.5.6. At TP12, observe the PRE RANGE TRIG output signal as specified in 3.5.6.

4.6.3.7 T2 output. - At TP13, observe the T2 output signal as specified in 3.5.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548046. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The range counter assembly is used in the Computer Range CP-888/VPS-2, part of Radar Set AN/VPS-2. The range counter supplies the receiving system of the radar with range displaced triggers used to position the receiver gates at the predicted range of the target. A delayed 7.5 MHz clock pulse signal is also provided for use in the computer circuits.

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6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

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Project No. 1285-A241

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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